Experiment 1 – Design Star and Snowflake Schema

Aim: To design Star Schema and Snowflake Schema for the **Jio Hotstar** OTT Streaming Platform using a suitable fact table and dimension tables.

Objective:

I. To understand the concept of Data Warehouse Schema.
II. To design a Star Schema for JioHotstar user watch activity.
III. To design a Snowflake Schema for JioHotstar user watch activity.

1. Analyze High-Concurrency Telemetry Data: To structure an analytical data warehouse model capable of capturing and aggregating millions of real-time streaming events during high-concurrency live events like IPL matches.

2. **Model Streaming Engagement Metrics:** To design fact-dimension relationships for evaluating core user engagement metrics, such as video completion rate, drop-off rates, and average view duration per session (Watch_Duration_mins).

3. Track Revenue & Subscription Performance Across Cohorts: To model transactional relationships between subscription plans (DIM_PLAN), payment channels, and user demographic segments for dynamic lifetime value (LTV) calculation.

Theory:

1. Data Warehouse Schema Concepts

➔ A Data Warehouse Schema is the logical organization and structure of data in a relational database or data warehouse.

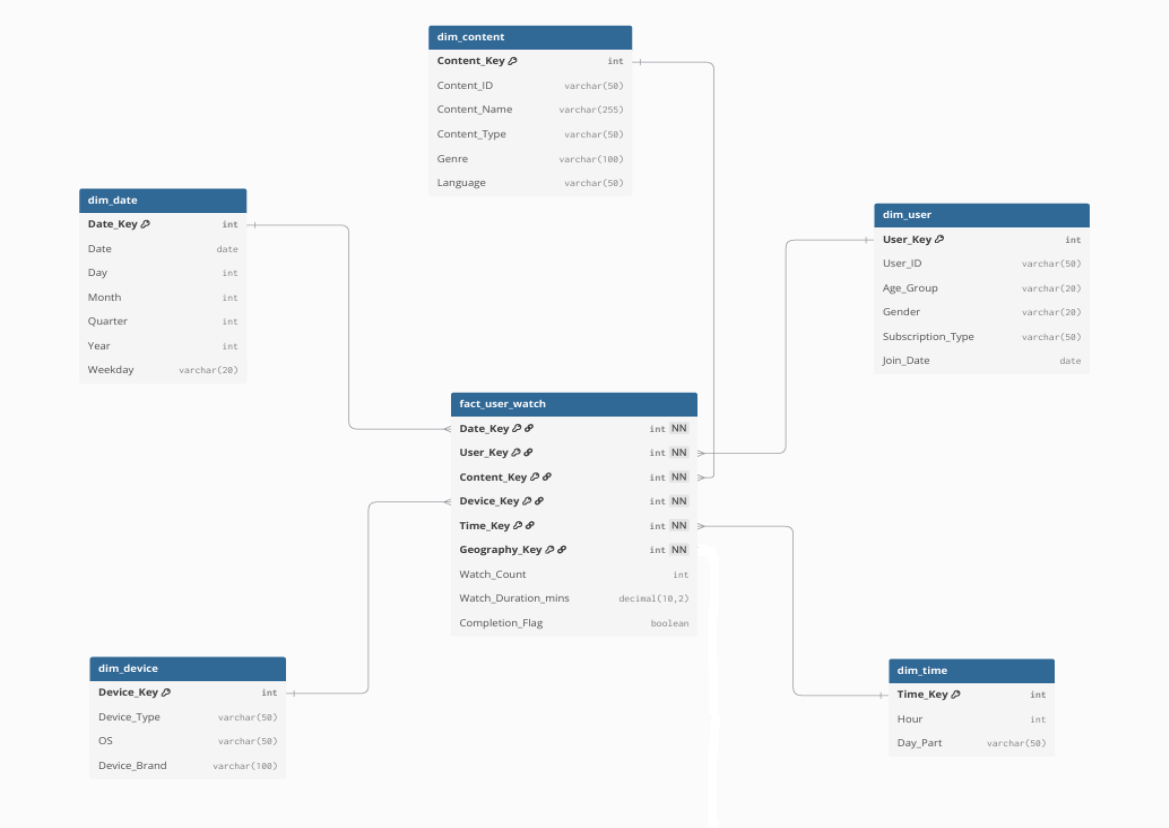
In a platform like **JioHotstar**, millions of user telemetry records—**such as video playback sessions, subscriber transactions, and live sports viewership**—are transformed into Fact and Dimension tables for business decision-making.

2. Star Schema

➔ A **Star Schema** is the simplest and most common data warehouse design pattern. In this architecture, a central **Fact Table** is directly connected to multiple non-normalized **Dimension Tables** in a single star-like join pattern.

- **Fact Table (fact_user_watch):** Stores numeric, quantitative measurements and foreign keys that link to surrounding dimension tables.
- **Dimension Tables:** Contain descriptive, context-rich attributes used for filtering, grouping, and slicing analytical queries (e.g., user demographics, content details, device types).

Structure No 1:



Fact Table:

fact_user_watch

Column	Description
Date_Key	Primary Key
User_Key	Foreign Key
Content_Key	Foreign Key
Device_Key	Foreign Key
Time_Key	Foreign Key
Watch_Count	Number of watch sessions
Geography_Key	Total sales amount

Dimension Tables:

dim_date

Column	Description
Date_Key	Primary Key
Date	Calendar date
Day	Day of the month

Month	Month number
Quarter	Quarter of the year

dim_content

Column	**Description**
Content_Key	Primary Key
Content_ID	Unique content identifier
Content_Name	Title of movie, TV show, or live sports
Content_Type	Type of content

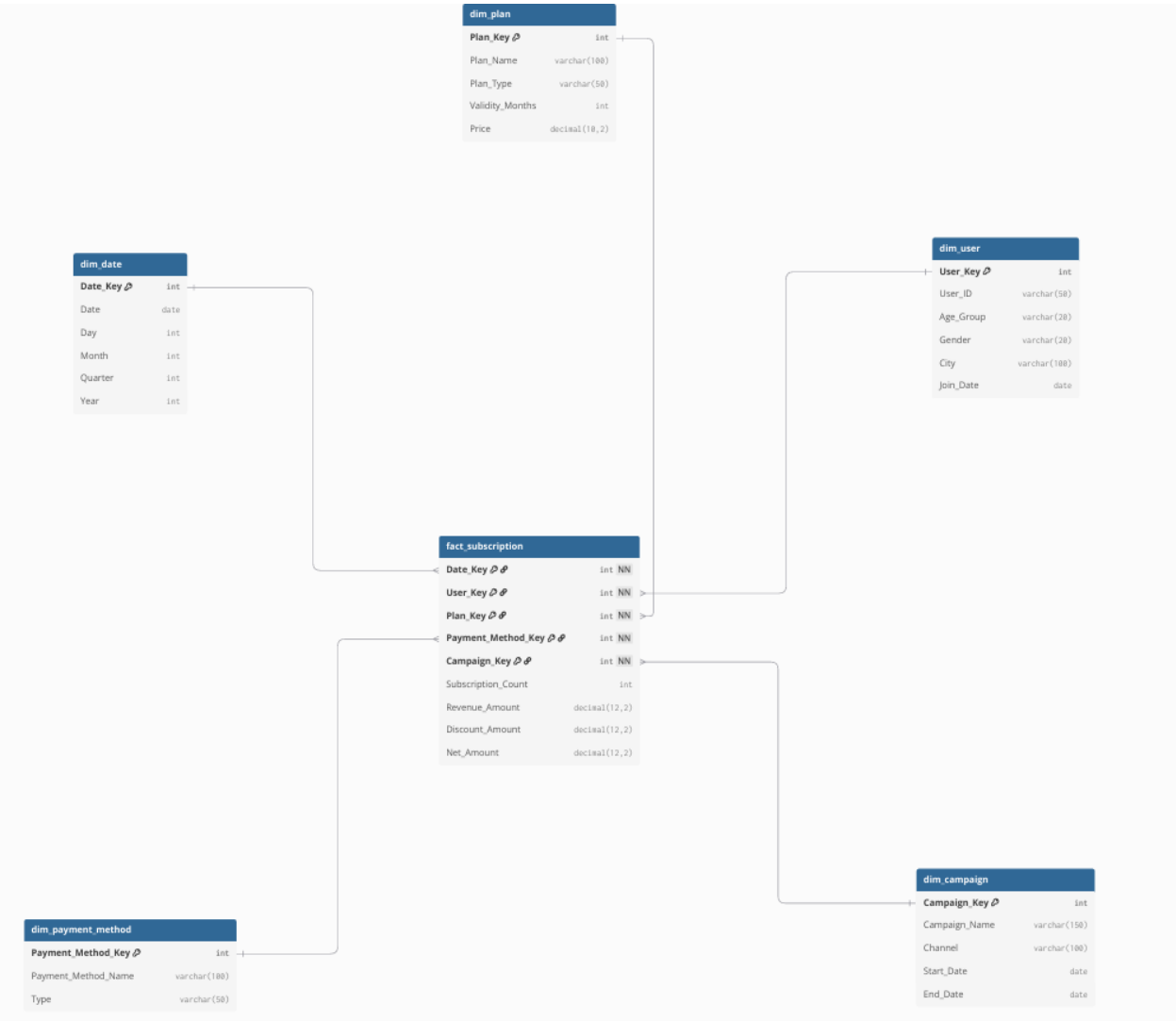
dim_device

Column	**Description**
Device_Key	Primary Key
Device_Type	Device type (Mobile, Smart TV, Web)
OS	Operating System
Device_Brand	Hardware manufacturer

dim_time

Column	Description
Time_Key	Primary Key
Hour	Hour of the day (0 to 23)
Day_part	Time segment (Morning, Prime Time, etc.)

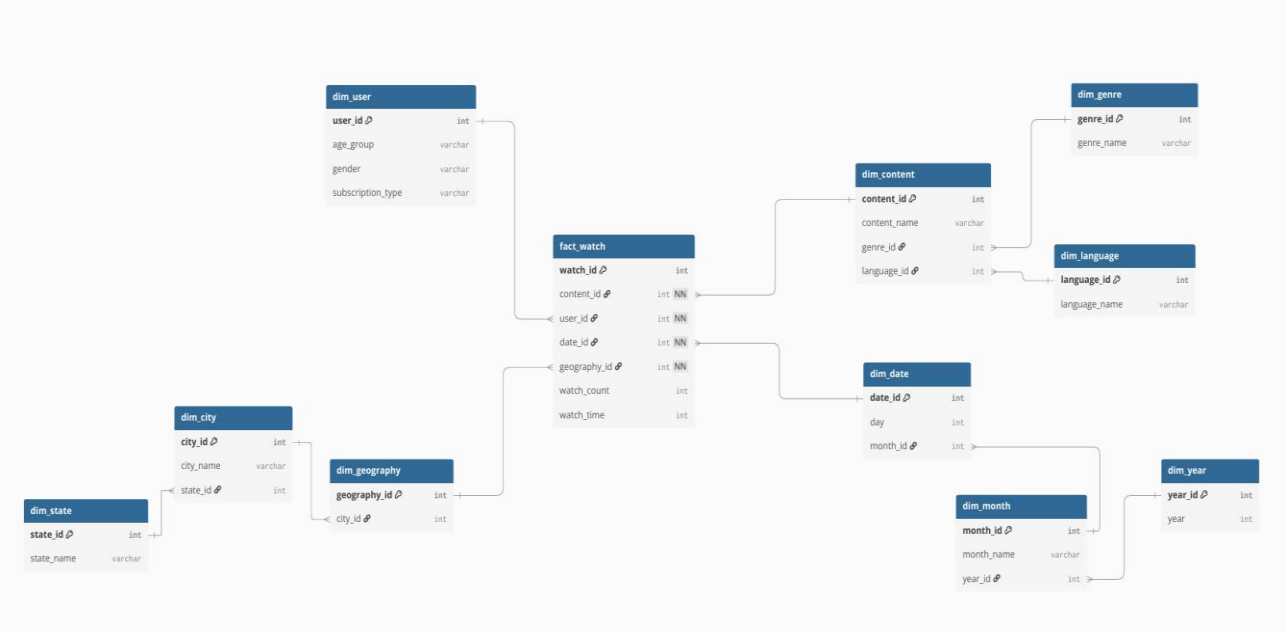
Structure No 2:



3. Snowflake Schema:

➔ A snowflake schema is a multi-dimensional data model that extends the star schema by breaking down dimension tables into normalized sub-dimension tables. This hierarchical branching layout resembles a snowflake shape on an entity-relationship diagram, helping to optimize storage and manage complex data relationships.

Structure:



Main Differences

Feature	Normalized	Denormalized
Data Redundancy	Data Redundancy	It can lead to redundancy
Write Operations	Write Operations	It can be slower due to redundancy
Read Operations	Read Operations	It is faster due to fewer table joins
Data Consistency	Data Consistency	It can lead to inconsistencies

Result:

Thus, the Star Schema and Snowflake Schema for the JioHotstar OTT Streaming Platform were successfully designed by identifying the User Watch Fact Table and the corresponding Dimension Tables.